Emanuel André Medina Araujo

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Education

Technical University of Munich — Oct. 2026 – Present
Master of Science in Informatics — *Munich, DE*

Technical University of Munich — Oct. 2020 – Aug. 2026
Bachelor of Science in Informatics (Computer Science), Application Area in Physics — *Munich, DE*

Technical University of Munich — Oct. 2024 – Sep. 2025
Parallel studies in Physics (B.Sc. programme, not completed) — *Munich, DE*

Herzog-Christoph-Gymnasium — Sep. 2012 – Sep. 2020
Abitur, Awarded MINT-EC certificate with distinction — *Beilstein, DE*

Experience

Working Student Software Engineer — Nov. 2024 – Present
Vector Informatik GmbH - VSceneCreator (3D OpenDRIVE Editor) — *Munich, DE*

- Helping take VSceneCreator (Unity/C#) from its first alpha (0.1) to the stable 1.0 release (Dec. 2026), with 43 merged PRs.
- Reduced edit-mode switch times by up to 70% (previously about 2 min for a 40 MB file) by restructuring the editor's base model to lazy-load non-essential and non-visible objects.
- Cut CI time for editor-only pull requests from 29.8 to 18.8 min and from 43.9 to 33.7 min on two build nodes (Jenkins) by skipping unrelated packaging, reusing module results and retrying flaky tests in place.
- Built a polyline trajectory editing mode with load and save for OpenSCENARIO (driving-scenario standard): data model, controllers, inspectors, validation issues, and a lossless round-trip of optional XML attributes.
- Designed and implemented geometry simplification that detects consecutive line and arc segments with matching heading and curvature in OpenDRIVE road networks and merges them through single or bulk fixes, with undo/redo and live validation.
- Brought the entire VSceneCreator application to Linux by abstracting platform-specific dialogs and file operations and making three native C/C++ libraries build and load there, with a Conan setup for both platforms.

Working Student Software Engineer — Mar. 2021 – Oct. 2024
Vector Informatik GmbH - DYNA4 (Simulation environment for virtual driving tests) — *Munich, DE*

- Merged 218 pull requests across nine releases (6.0 to 9.1) in Java and Eclipse RCP, working directly with Product Management.
- Extended the run-package export from scenarios to tasks, refactoring the simulation runner to export instead of simulate.
- Built reference analysis so users save only the files a database sync references; co-developed warning markers and a save quick fix for unsaved parameter files, removing a deadlock risk by no longer querying the UI from background jobs.
- Reduced manual QA effort with 48 test-focused pull requests (SWTBot UI tests, JUnit), test code in 120 of 218 pull requests overall, and a custom JUnit formatter that links Jenkins results to Jira tickets through Xray.

Projects

LoveNeverDies | *Legal Loves Tech Hackathon – Bavarian Ministry of Justice Challenge* — Aug. 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guiding users through inheritance deadlines via a 141-node legal decision tree.
- Designed a zero-trust architecture in which the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM, and a Shamir-secret-shared vault.
- Verified row-level security against real Postgres migrations with 130 Vitest files and multi-user Playwright E2E tests.

Learning-Based Inverse Kinematics | *Bachelor Thesis* — Jun. 2024 – Nov. 2024

- Compared learning-based IK solvers (MLP, LSTM, hybrid, PPO) for 2- and 3-DOF manipulators in PyTorch on a 10k-sample test set; a simple MLP with direct-angle prediction consistently outperformed the more complex models.

Ironman | *3.8km swim, 180km bike, 42.2km run; Overall Time : 14:21:26h* — Oct. 2022 – Aug. 2024

- Completed a full Ironman triathlon through disciplined training and a goal-oriented race plan.

Technical Skills

Languages: C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB | **Standards**: OpenDRIVE, OpenSCENARIO
Frameworks: React, React Native, Unity, PyTorch, Eclipse RCP, EMF, JavaFX, Supabase/PostgreSQL, WebRTC
Developer Skills: Git, Linux, GitHub Actions, Jenkins, Conan, SWTBot, JUnit, Playwright, Vitest, Rider, VS Code, VIM Motions, Jira